THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 10 (2020/2021)

1. (a) Let $P(x) = -0.01x^2 + 550x - 10,000$ be the profit function for a monopolist. Find the value of x that maximizes the profit and determine the corresponding profit. estimates that for each $K2$ price increase, 10 fewer loaves of bread will be sold. What cost will maximize the profit?

   (b) A printer needs to make a poster that will have a total area of $200m^2$ and will have $1m$ margins on the sides, a $2m$ margin on the top and a $1.5m$ margin on the bottom. What dimensions of the poster will give the largest printed area?

   (c) A window is being built and the bottom is a rectangle and the top is a semicircle. If there is $12m$ of framing materials what must the dimensions of the window be to let in the most light?

2. Find the following integrals:

   (a)
$$\int (x^{\frac{1}{3}} + x^{-\frac{1}{3}})\ dx$$

   (b)
$$\int (2x+3)x^2\ dx$$

   (c)
$$\int \sqrt{x}(x+2)\ dx$$

   (d)
$$\int \left(x + \frac{1}{x}\right)^2\ dx$$

   (e)
$$\int (\sqrt{x}+2)^2\ dx$$

   (f)
$$\int \left(3 + \frac{\sqrt{x}+6x^3}{x}\right)\ dx$$

   (g)
$$\int \frac{(2x+1)^2}{\sqrt{x}}\ dx$$

3. (a) The gradient of a curve is given by $f'(x) = x^2 - 3x - \frac{2}{x^2}$. Given that the curve passes through the point $(1,1)$, find the equation of the curve in the form $y = f(x)$.

   (b) Given that
$$y^{\frac{1}{2}} = x^{\frac{1}{3}} + 3:$$

      i. Show that
$$y = x^{\frac{2}{3}} + Ax^{\frac{1}{3}} + 3,$$
      where A and B are constants to be found.

      ii. Hence find
$$\int y\ dx$$

4. (a) Evaluate each of the following:

i.

$$\int_1^8 (x^{-\frac{1}{3}} + 2x - 1)\ dx$$

ii.

$$\int_1^3 \left(\frac{x^3 + 2x^2}{x}\right)\ dx$$

iii.

$$\int_1^4 (\sqrt{x} - 3)^2\ dx$$

iv.

$$\int_3^6 \left(x - \frac{3}{x}\right)^2\ dx$$

v.

$$\int_0^1 x^2 \left(\sqrt{x} + \frac{1}{x}\right)\ dx$$

vi.

$$\int_1^4 (x^{\frac{1}{2}} - 4)(x^{-\frac{1}{2}} - 1)\ dx$$

(b) Find the area between the curve with equation $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$ in each of the following:

i. Find the point at which the gradient of the curve

$$f(x) := 3x^3 - 2x + 2; \qquad a = 0, b = 2.$$

ii. Find the point at which the gradient of the curve

$$f(x) := \sqrt{x} + 2x; \qquad a = 1, b = 4.$$

iii. Find the point at which the gradient of the curve

$$f(x) := \frac{8}{x^3} + \sqrt{x}; \qquad a = 1, b = 4.$$

(c) Sketch the graph of $y = x(x^2 - 4)$, and hence find the area of the region, above the x-axis, bounded by the curve and the x-axis.

(d) Find the area of the finite region between the curve with equation $y = (3 - x)(1 + x)$ and the x-axis.

5. (a) Sketch each of the following and find the area of the finite region or regions bounded by the curves and the x-axis.

i.

$$f(x) = x(x + 2)$$

ii.

$$f(x) = (x + 1)(x - 4)$$

iii.

$$f(x) = x(x + 3)(x - 3)$$

iv.

$$f(x) = (x - 2)x^2$$

v.

$$f(x) = x(x - 2)(x - 5)$$

(b) In each of the following, sketch the pairs of functions on the same Cartesian plane. Hence find the area of the region(s) bounded by the two functions.

i.

$f(x) = 4x - x^2$ and $y = 3$.

ii.

$f(x) = 3x^2 - 2x$ and $y = 1 - 4x$.

iii.

$f(x) = x^2 + 1$ and $y = 7 - x$.

iv.

$f(x) = x^3 - 3x$ and $y = x$.